SESSION 4 · ECONOMIC IMPLICATIONS OF AI

How much can AI add to productivity and GDP?

Evidence for Latin America

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Duke University

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Today: two steps for 22 countries of Latin America and the Caribbean, with public data

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- Gain = GDP weight of the tasks AI makes cheaper × how much cheaper
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(b) Baumol's cost disease through structural transformation

- Sectoral weights are not fixed: employment and GDP shares respond to sectoral productivity
- Baumol (1967): the sectors that progress fastest shrink in weight; the stagnant ones grow
- Paradox: productivity growth converges to the growth of the sector without the technological improvement
- Structural transformation can matter due to non-linearities

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> **A historical lens.** US agriculture: 41% of workers in 1900, under 2% today, while its productivity grew faster than any other sector's

The bound is a product of four factors

TFP =

s_L
Labor share
Share of GDP paid to labor: the wage bill

×

e
Exposure
Share of that wage bill in tasks AI could perform

×

q
Profitability
Share of those tasks worth automating within ten years

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Employment by occupation and relative wages, ILOSTAT; task exposure by occupation, Eloundou et al. (2024)

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AGHION AND BUNEL (2024)
40%, three US experiments with generative AI (27% in Acemoglu)

Cost saving. Acemoglu uses 27%, the mean of two experiments (writing, customer support). Aghion and Bunel (2024) add the coding experiment (Peng et al. 2023, 56%): 40%. Coding is where adoption is fastest, and the gains appear fast and persist: 14% in the first month, about 25% from the second (Brynjolfsson, Li and Raymond)

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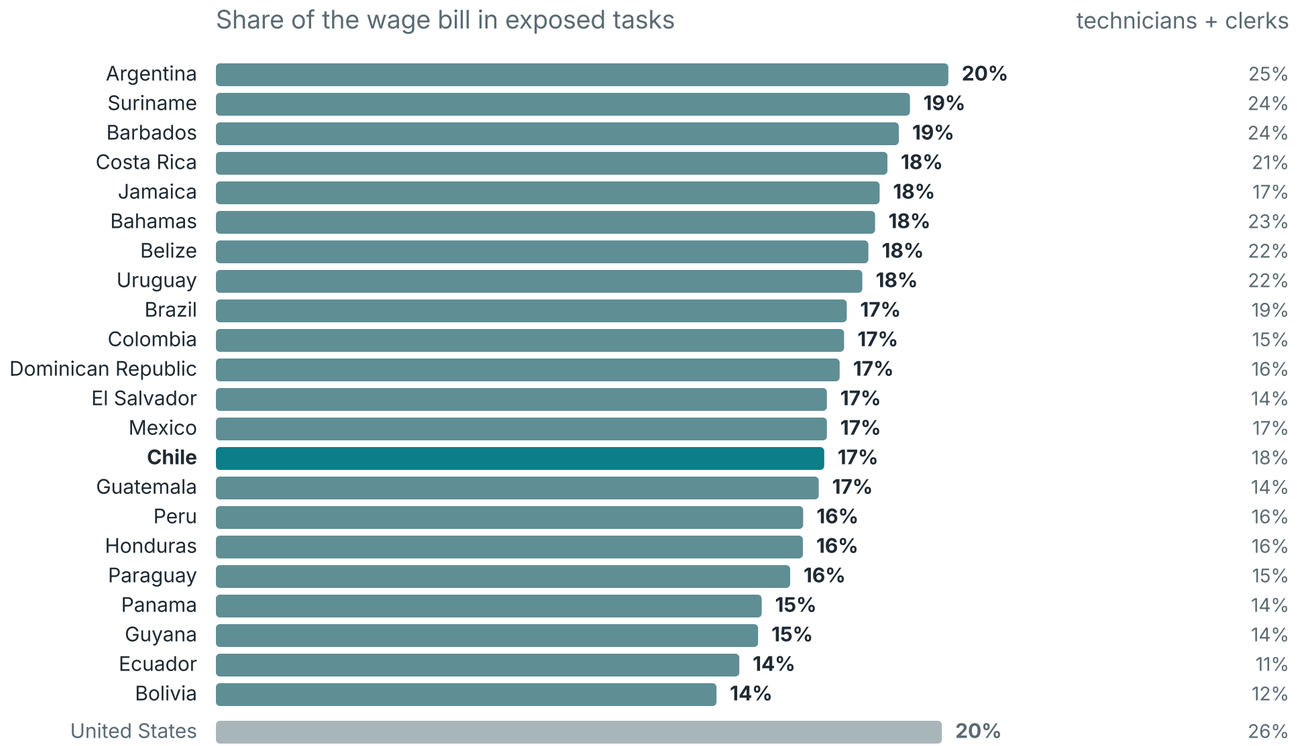
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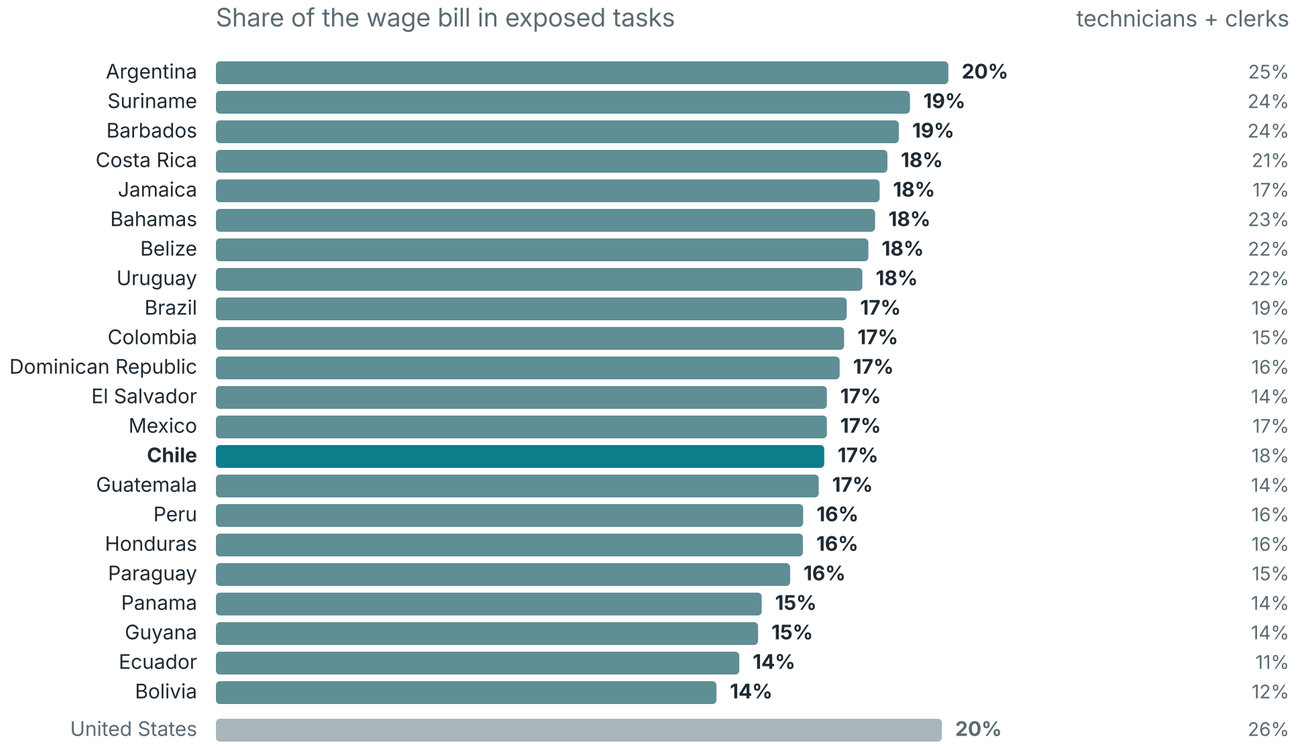
GDP = TFP / s_L

Production: Cobb-Douglas in labor and capital by sector; labor mobile; capital rented at a fixed price, so the capital-output ratio is constant (Acemoglu's multiplier, $1/s_L$: 1.75 in the US, 1.73 to 2.92 in the region). CES capital-labor as robustness (slide 9, A8)

14% to 20% of the wage bill is in exposed tasks; US: 20.0%

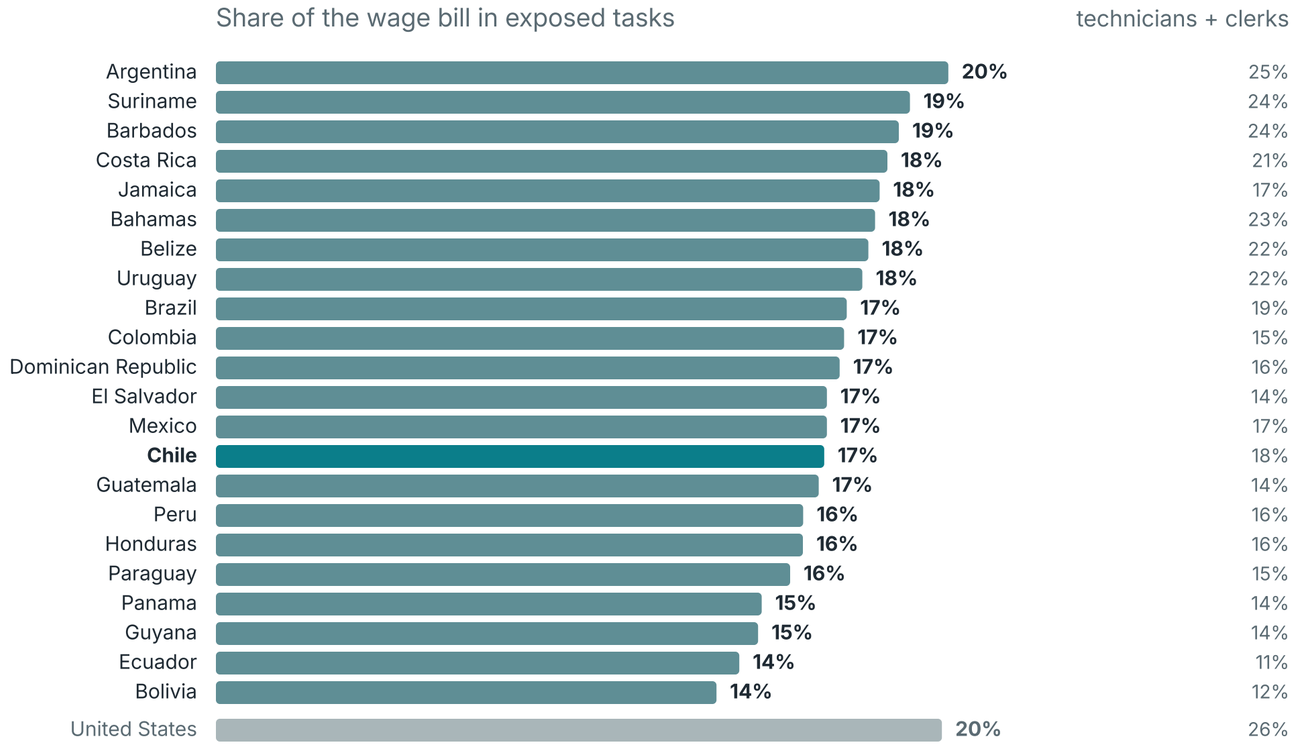


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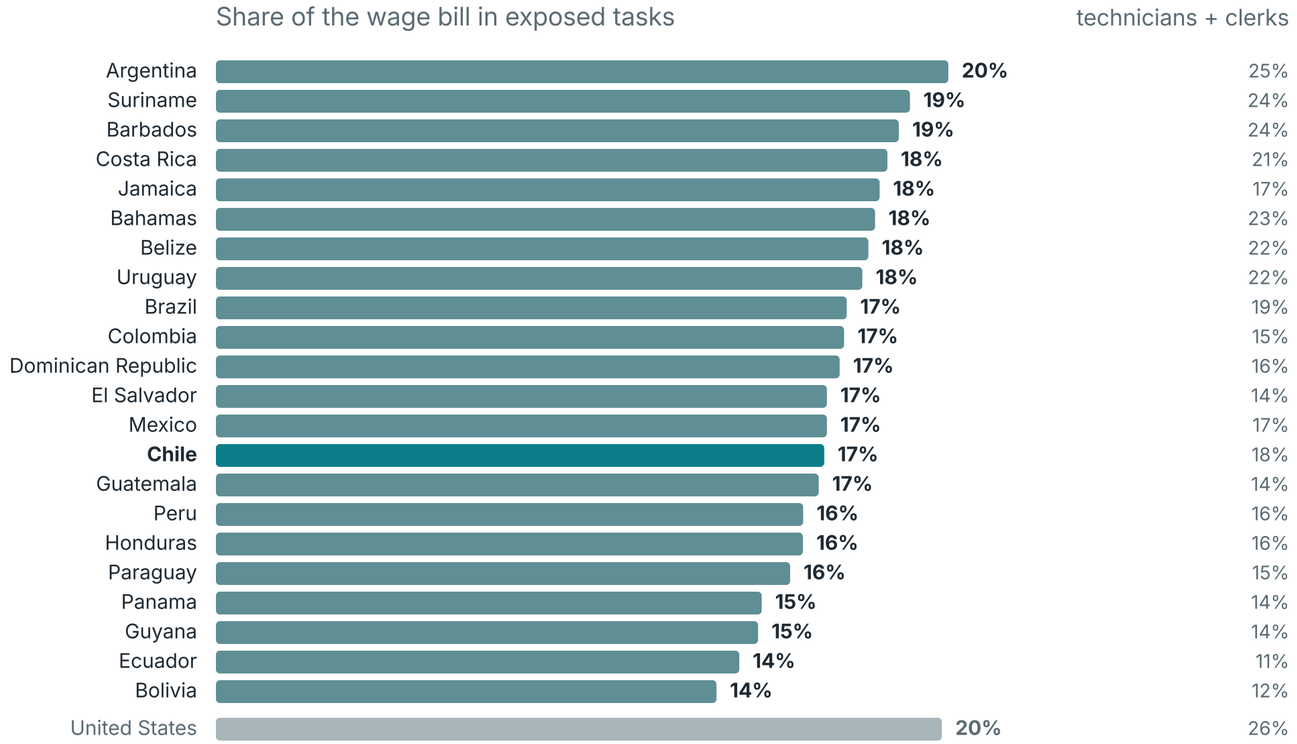
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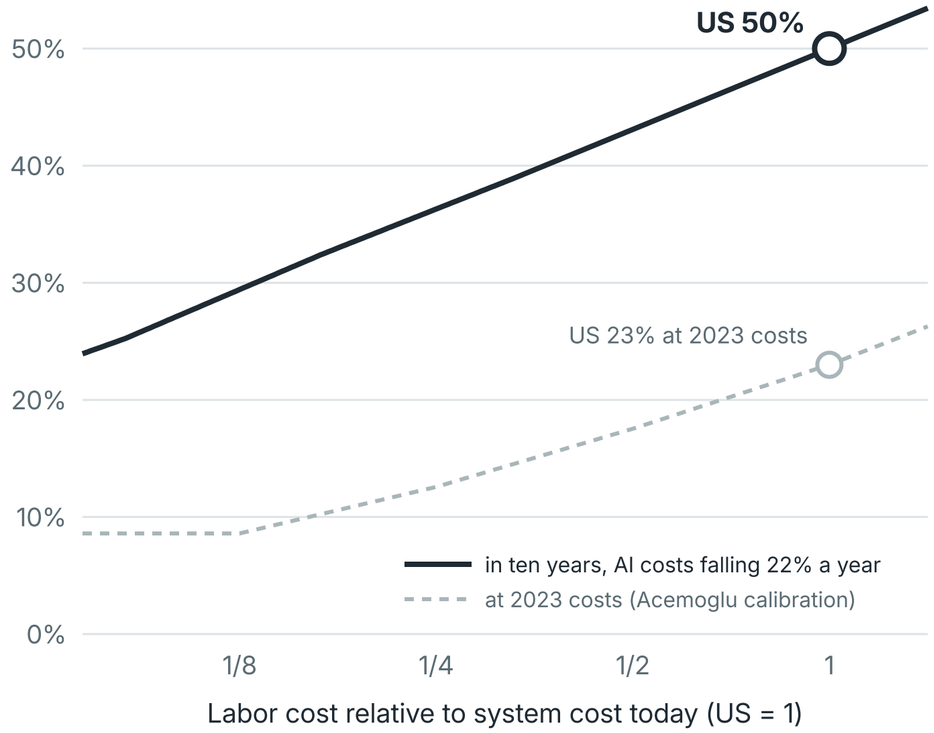
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Not Aghion and Bunel's 60%. Their US exposure is the IMF occupation index: occupations touched by AI, not tasks AI can do. The task-based measure is kept, so the benchmark is below their own 0.68 pp a year

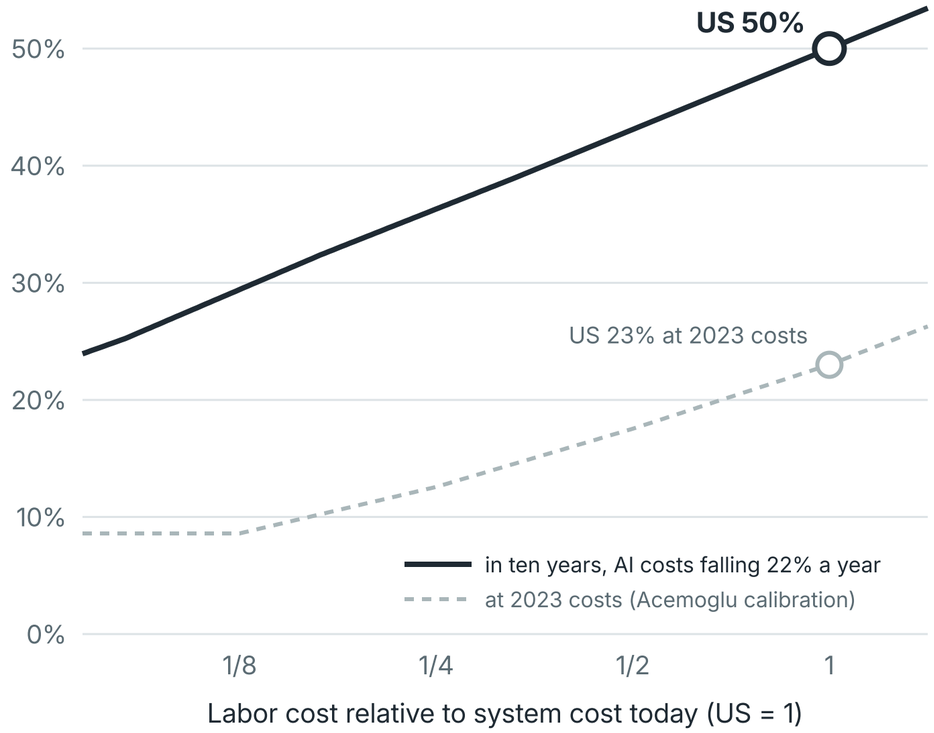
25% to 45% of exposed tasks pay off within ten years, versus 50% in the US

Exposed tasks worth automating (cost curve of Svanberg et al. 2024)

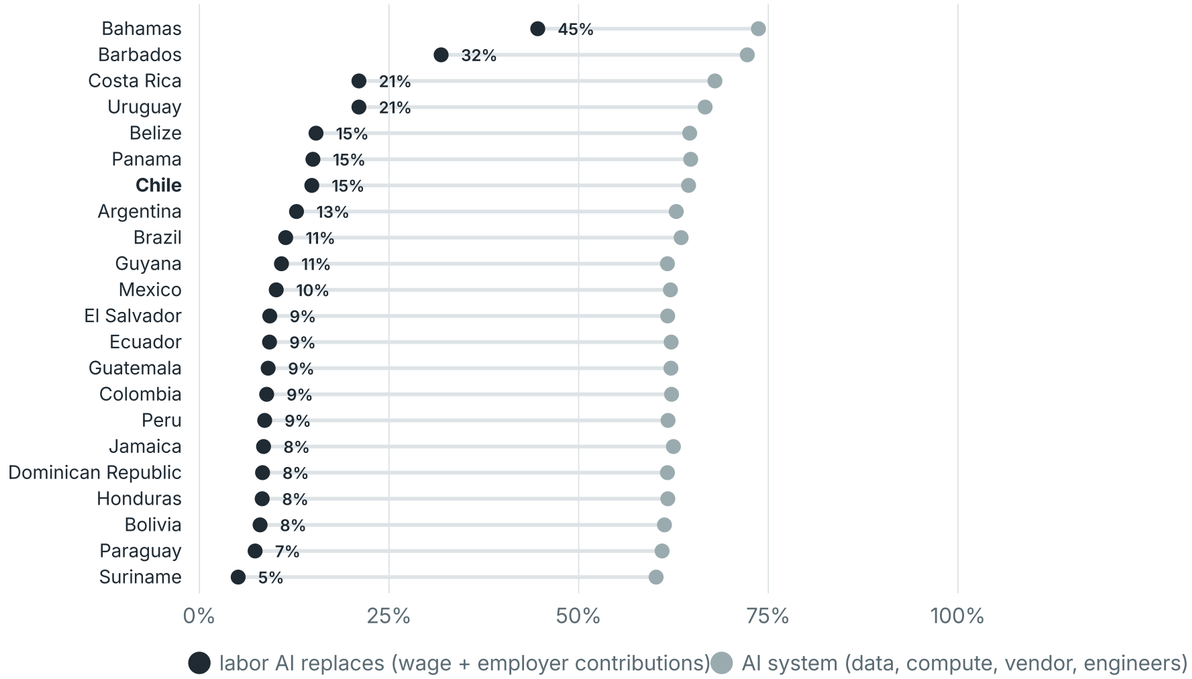


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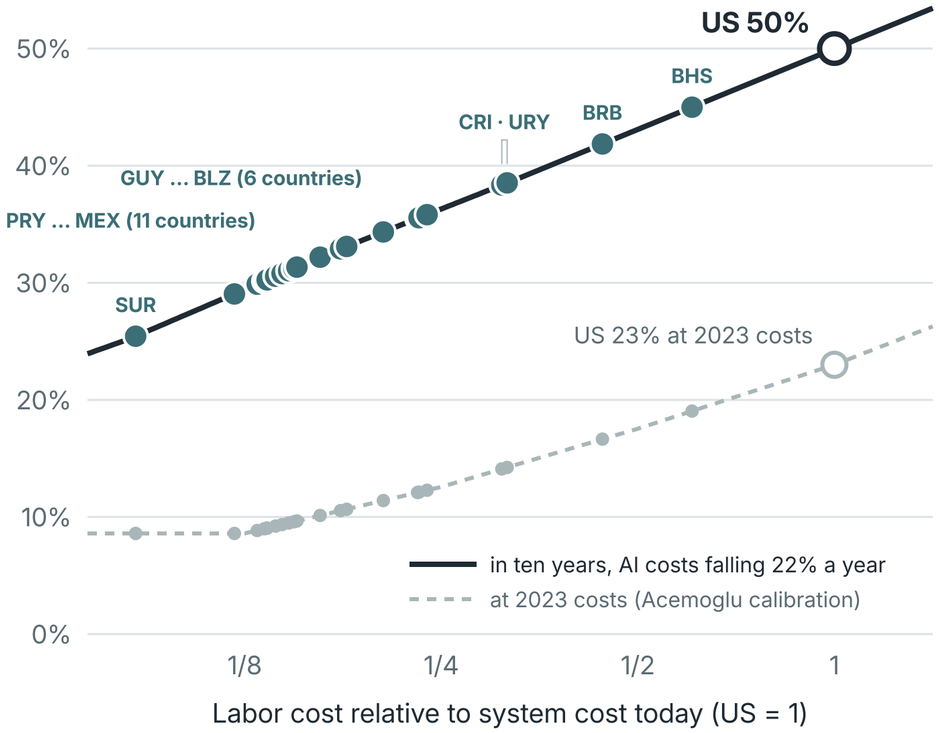
Cost relative to the United States (= 100%)



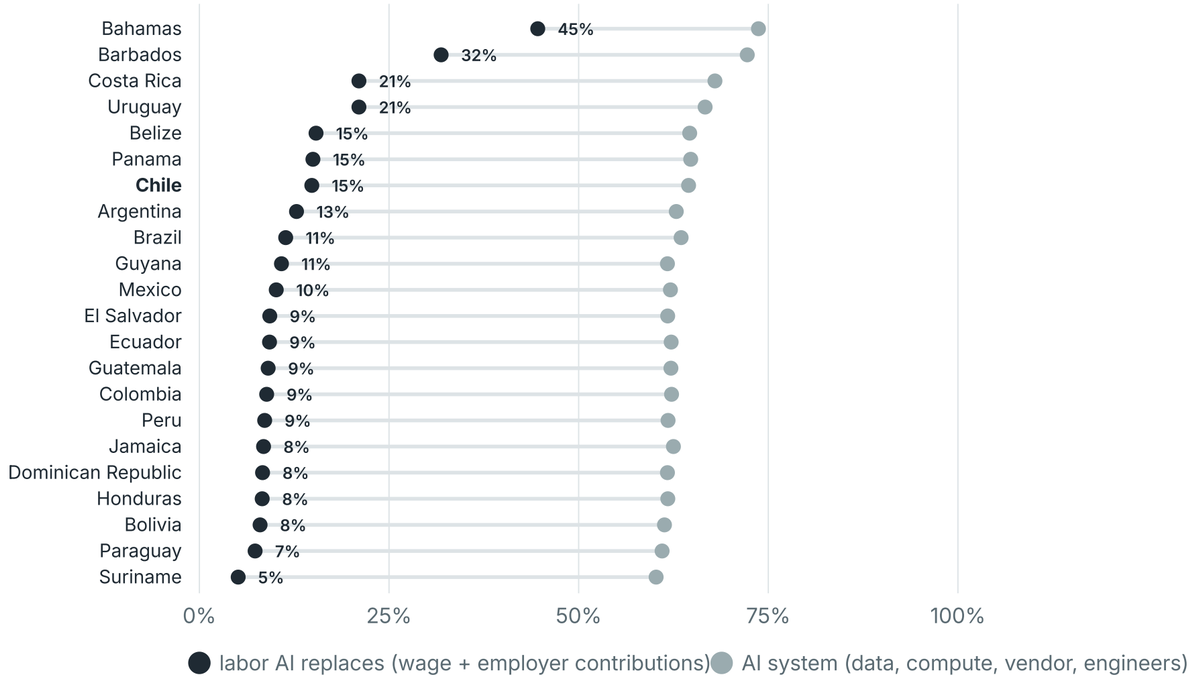
AI system: cost structure of Svanberg et al. (2024). 58% is paid at US prices (data, compute, external vendor); 42% is in-house engineers, at the earnings of professionals in each country relative to the US (ILOSTAT)

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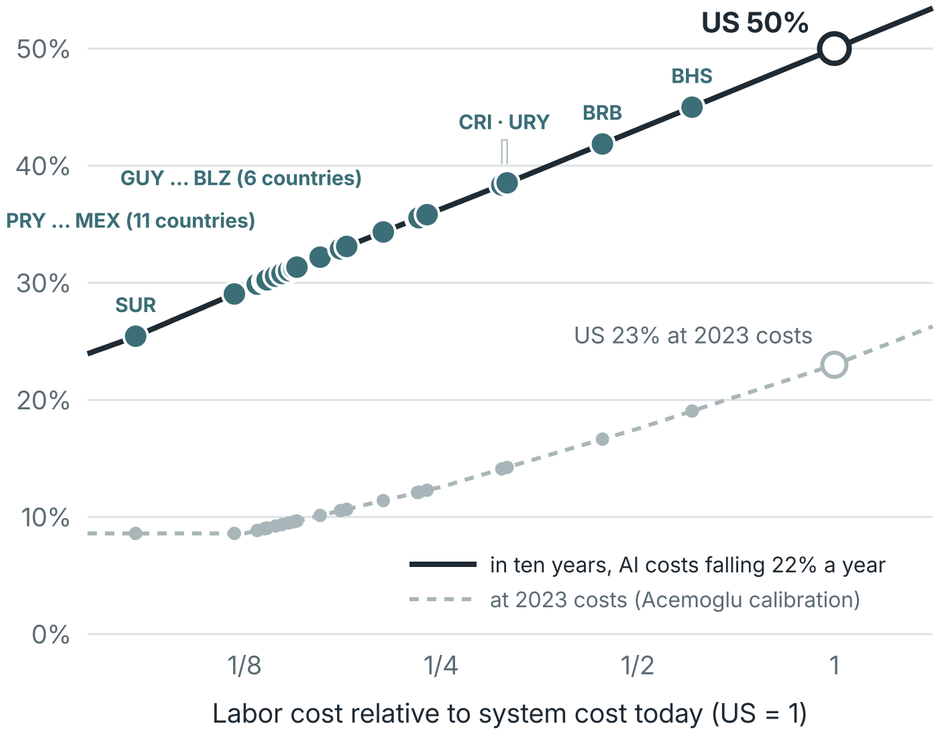
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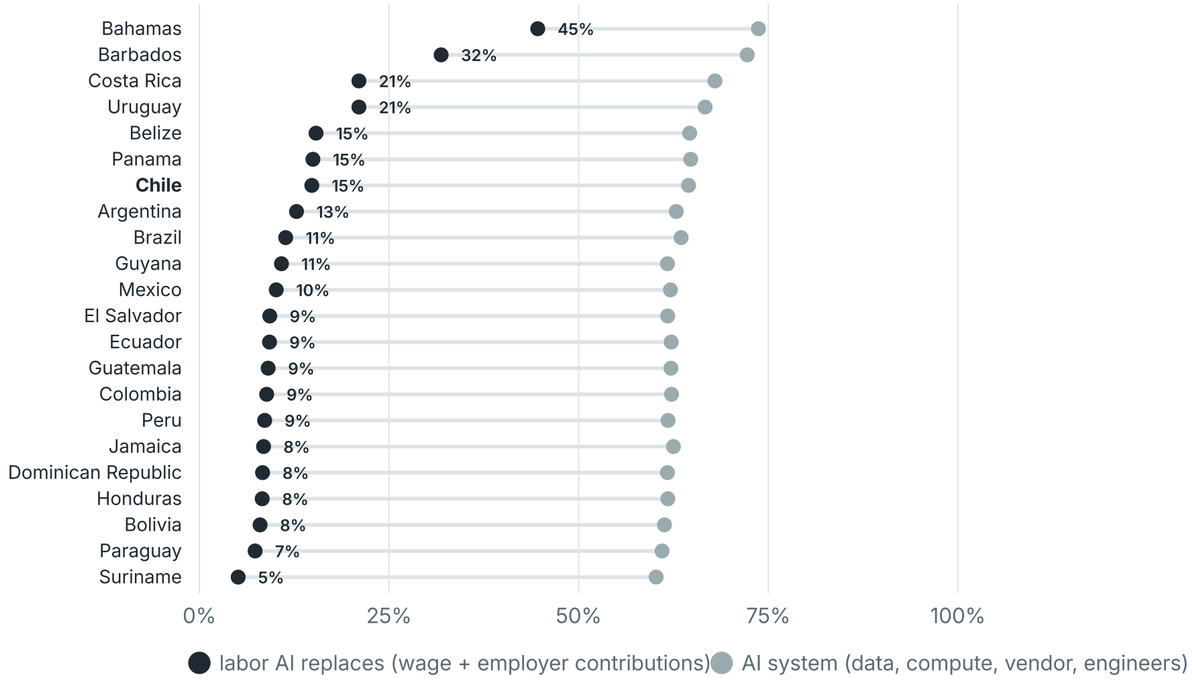
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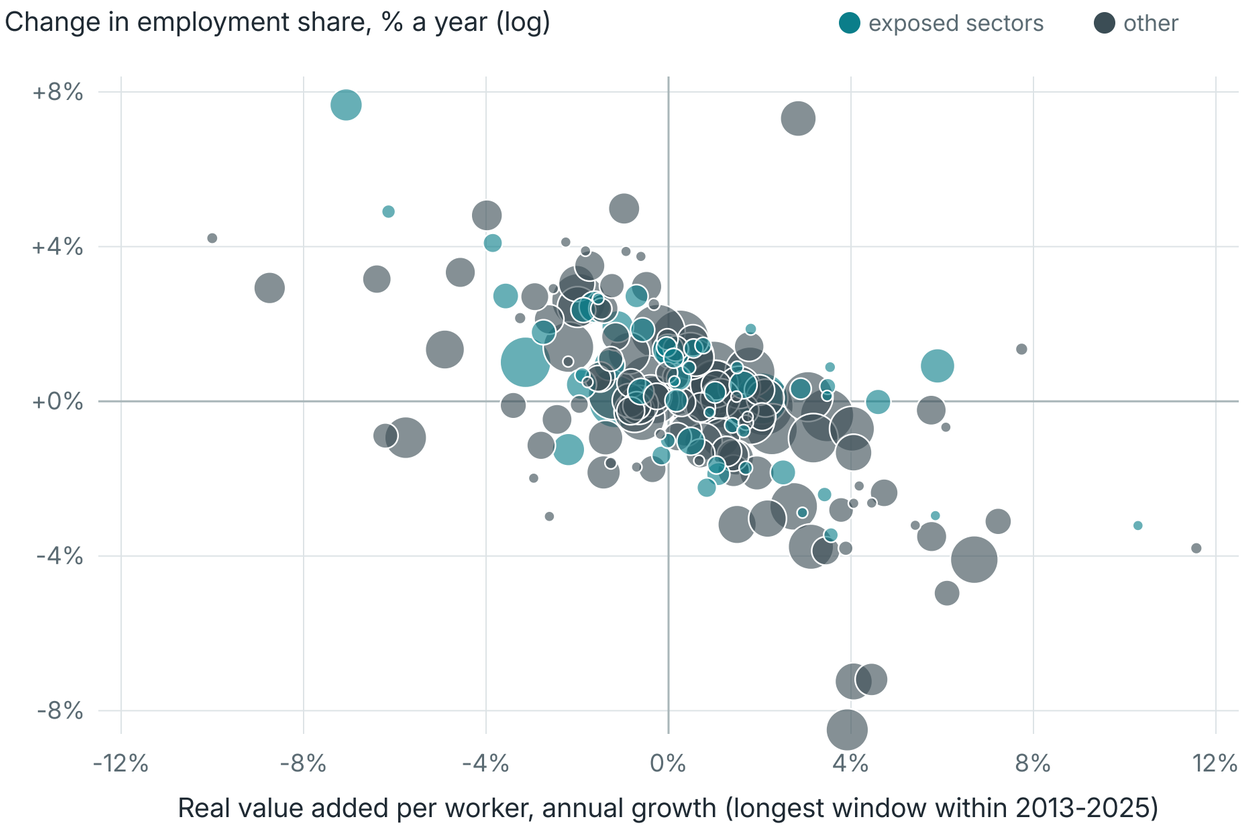
Aghion and Bunel let costs fall 22% a year in the same Svanberg model. How fast is AI moving?

× 2 every 2.5 years
computing power per dollar, GPUs 2006-2021 (Epoch AI 2022): costs falling 22% to 24% a year

÷ 280 in two years
price of AI at GPT-3.5 quality, $20 to $0.07 per million tokens, Nov 2022 to Oct 2024 (Stanford AI Index 2025)

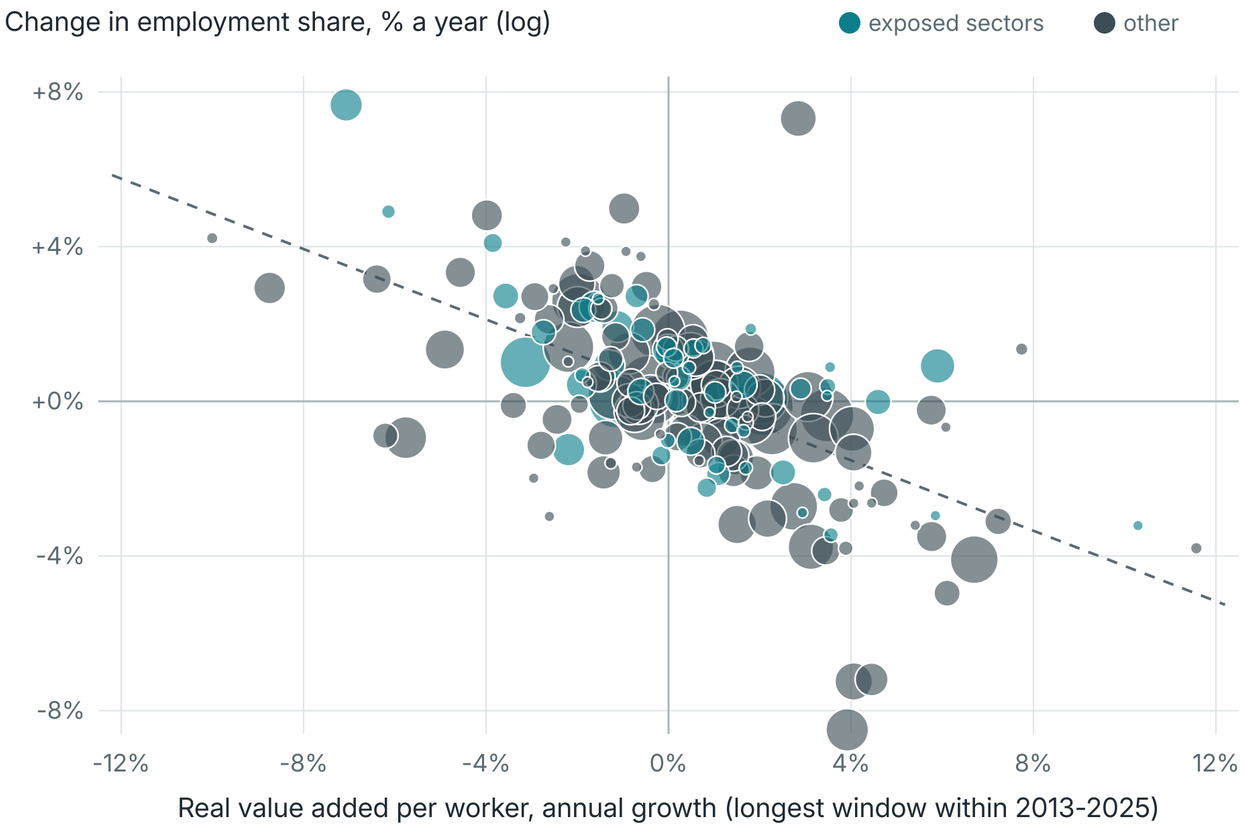
× 2 every 7 months
length of the software tasks AI completes on its own, in human-expert time, 2019-2025 (METR 2025)

Sectoral shares respond to productivity: Baumol cost disease



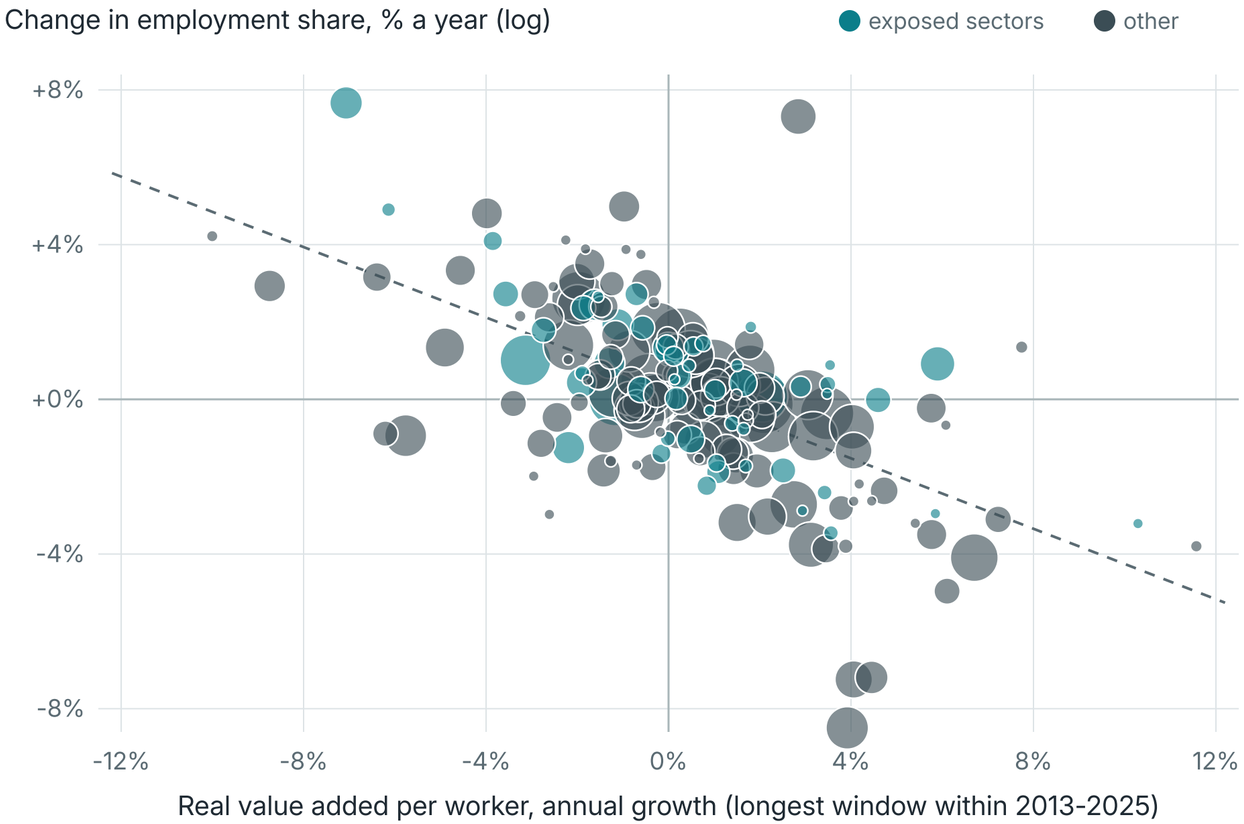
Observed data (CEPALSTAT, ILOSTAT), 18 countries × 11 sectors; area: employment share. No sectoral data: Paraguay, Suriname, Costa Rica, Jamaica
Correlation −0.79 pooled, −0.48 to −0.94 by country (without the oil boom in Guyana; −0.60 with it); 12 points off the chart

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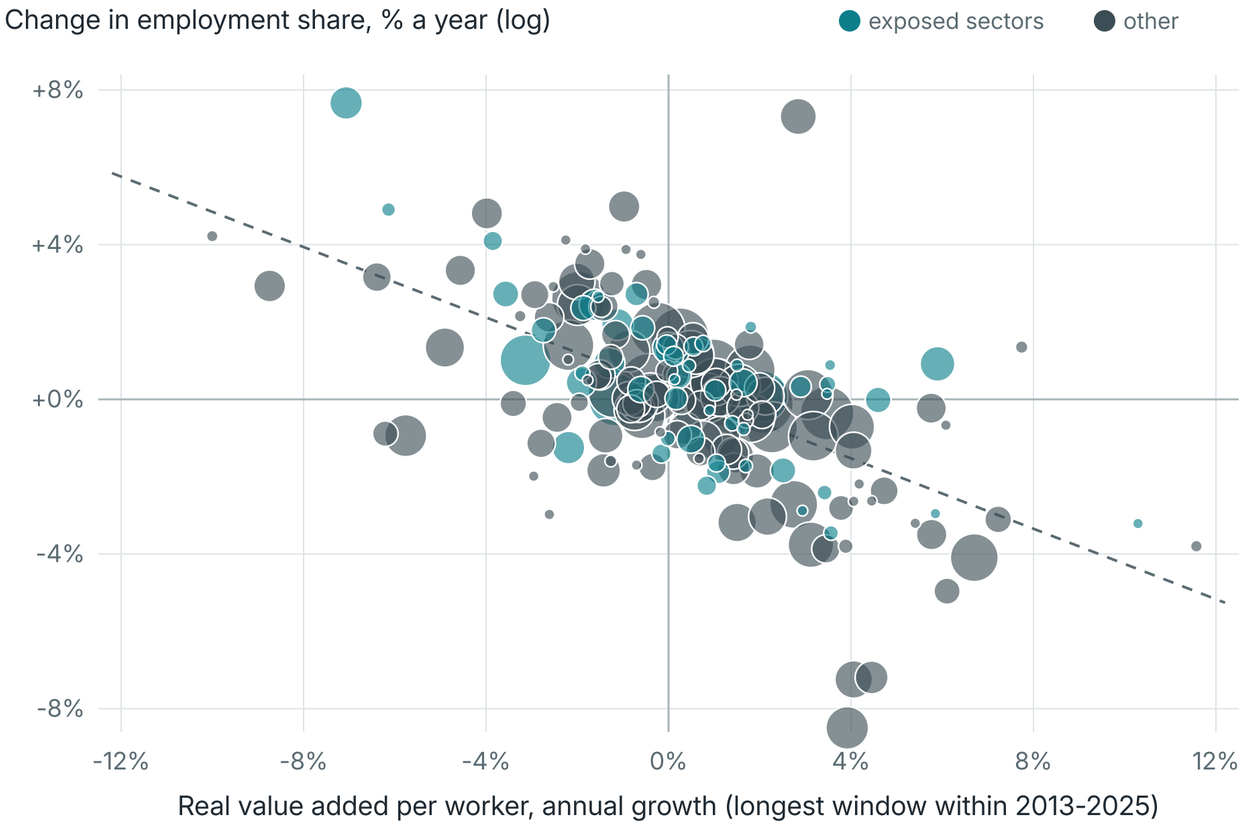
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$$s_i = \Omega_i \left(\frac{p_i}{E_n} \right)^{1-\sigma} C^{\varepsilon_i}, \quad \sum_i s_i = 1$$

- $\bar{c}_a$: subsistence in agriculture (Engel, income effect); C: the non-agricultural aggregate
- s_i: share of sector i in non-agricultural spending $E_n = \Sigma\, p_i c_i$; $\Sigma\, s_i = 1$ pins down C
- $\sigma = 0.35$: complements across sectors (Baumol, substitution effect)
- ε_i: income elasticities; finance and business 1.73, trade 1.38, mining 0.08 (manufacturing = 1)

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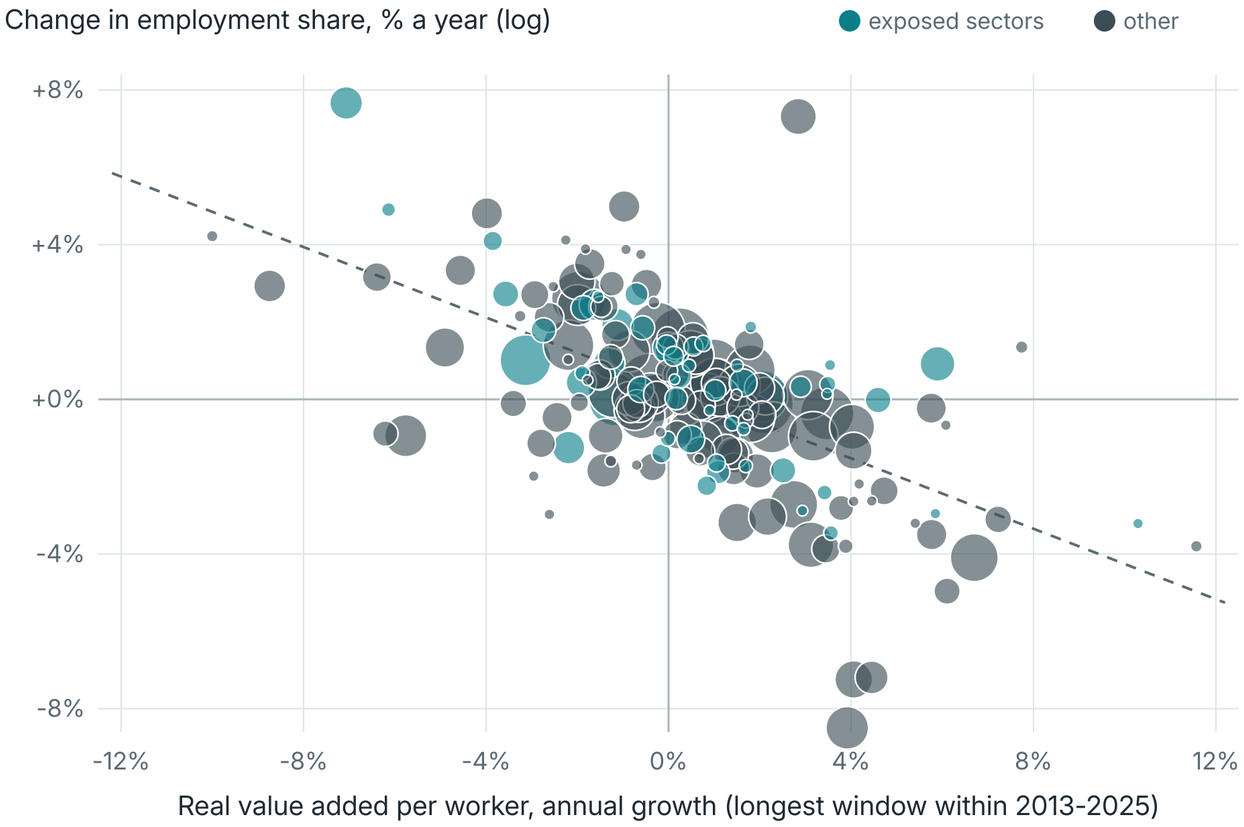
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- Each sector's productivity follows its observed 2013-2025 trend; the AI gain of part (a) enters sector by sector, on top of it
- With σ < 1 the fastest-growing sectors get cheaper, lose **nominal GDP weight** and release workers: the Hulten weights drift

Validation, in sample (13 countries where the model is run). Run over each country's own window (2013-2025 where available) with the observed productivity growth: predicted and actual changes in employment shares correlate 0.37 to 0.94 by country. No parameter is fitted to these paths; σ and ε_i come from the literature

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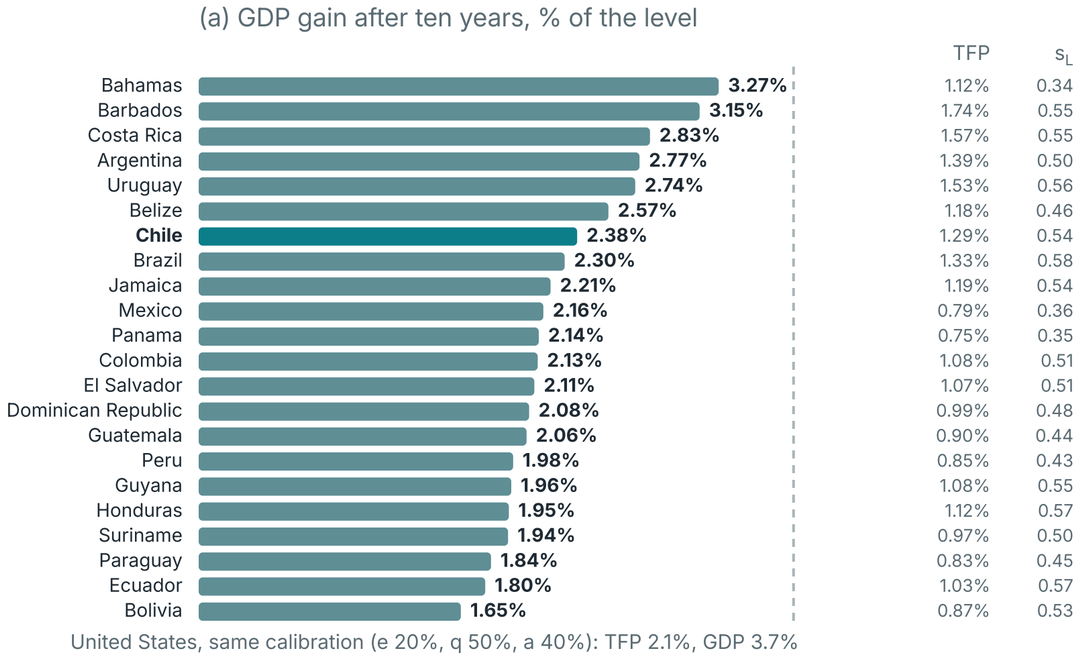
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This has happened before: agriculture, then industry, now services

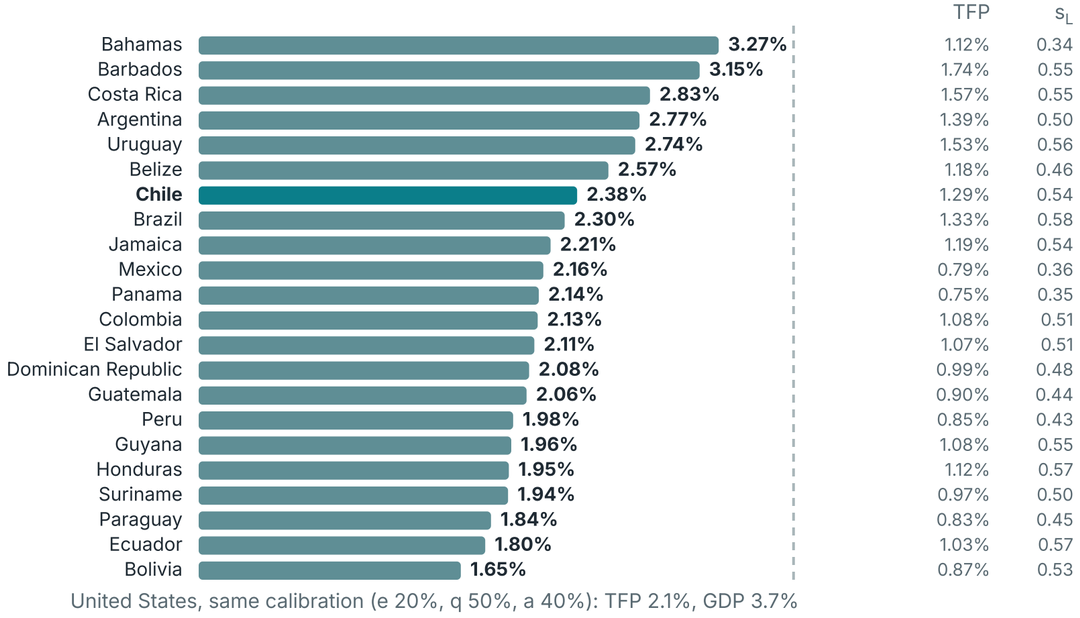
44% to 87% of the US effect; Baumol erodes up to 23% of it in twenty years



(a) A level gain cumulated over ten years, not a growth rate: GDP 1.65% (Bolivia) to 3.27% (Bahamas), 0.16 to 0.33 points of annual growth; TFP 0.7% to 1.7%. The labor share scales TFP and cancels in GDP = e × q × a

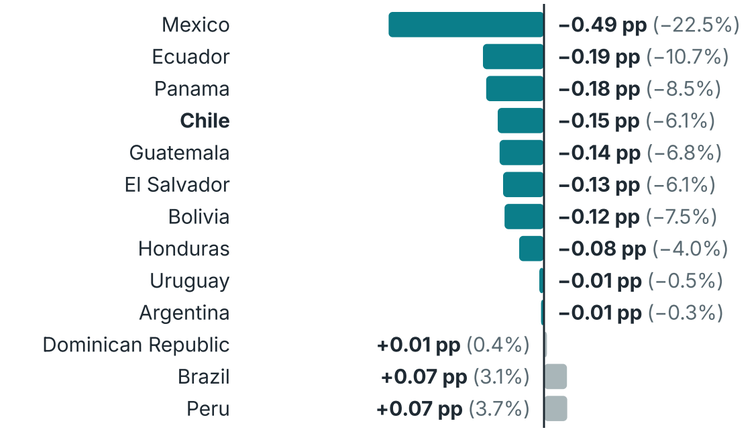
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(a) GDP gain after ten years, % of the level



United States, same calibration (e 20%, q 50%, a 40%): TFP 2.1%, GDP 3.7%

(b) Change in the GDP gain with 2042 instead of 2022 sectoral weights, pp of GDP



Not computed for 9 countries (data window or model range; list in the notes)

Chile: 2.38% with 2022 weights, 2.23% with 2042 weights: −0.15 pp (−6.1% of the gain)

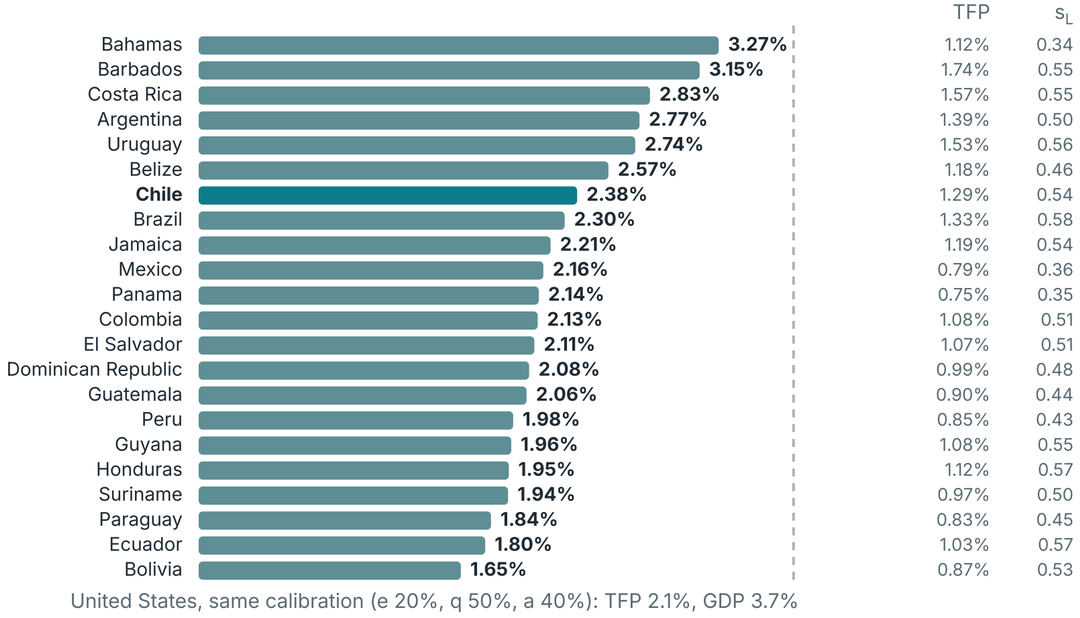
Twenty years of the observed sectoral trends; the AI gain itself stays the ten-year bound of part (a)

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(b) The same GDP gain with the sectoral weights the trends produce by 2042: −0.49 pp (Mexico) to +0.07 pp (Peru). It shrinks where the sectors with the largest AI gains are fast-productivity ones that lose weight; it grows where they are laggards that gain weight (Brazil, Peru, Dominican Republic)

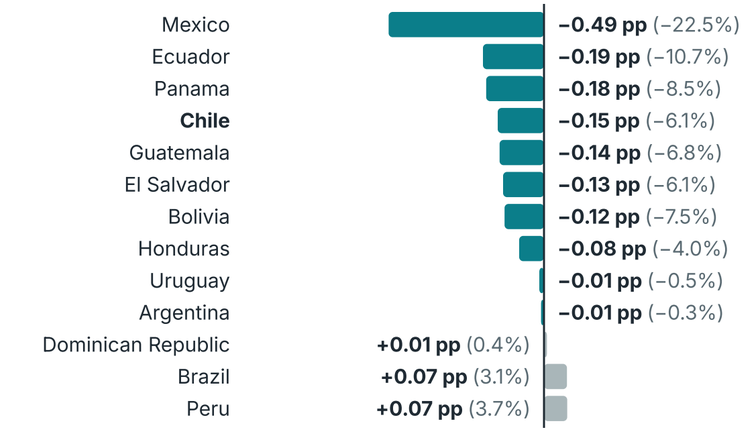
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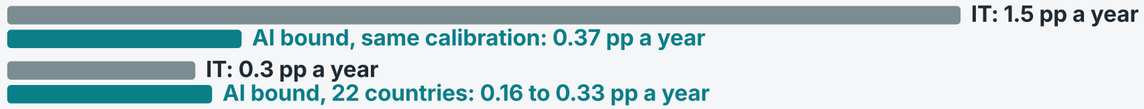


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Chile: 2.38% with 2022 weights, 2.23% with 2042 weights: −0.15 pp (−6.1% of the gain)
Twenty years of the observed sectoral trends; the AI gain itself stays the ten-year bound of part (a)

(a) A level gain cumulated over ten years, not a growth rate: GDP 1.65% (Bolivia) to 3.27% (Bahamas), 0.16 to 0.33 points of annual growth; TFP 0.7% to 1.7%. The labor share scales TFP and cancels in GDP = e × q × a

(b) The same GDP gain with the sectoral weights the trends produce by 2042: −0.49 pp (Mexico) to +0.07 pp (Peru). It shrinks where the sectors with the largest AI gains are fast-productivity ones that lose weight; it grows where they are laggards that gain weight (Brazil, Peru, Dominican Republic)

For scale: the IT revolution, 1995-2004 points of annual growth; IT from growth accounting, AI from the bound of part (a) spread over ten years

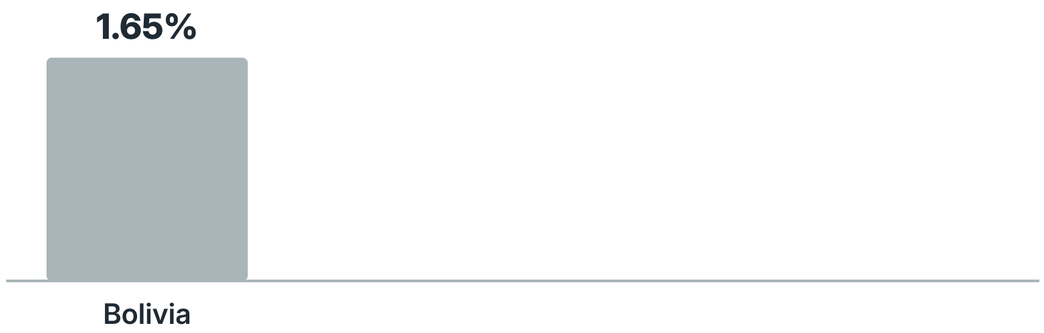


Sources: Byrne, Oliner and Sichel (2013), Table 1, nonfarm business, 1995-2004; Jorgenson and Vu (2007), Latin America, 1995-2004 (0.29 in 1995-2000, 0.30 in 2000-2004)

Bahamas vs Bolivia: 60% of the gap is profitability, 40% exposure

GDP bound in ten years, % of GDP: from the lowest country to the highest

GDP = e × q × a; a = 40% everywhere and the labor share cancels, so the gap splits exactly between e and q



Behind q: exposed-occupation labor costs 45% of the US level in Bahamas, 8% in Bolivia; the AI system costs 74% and 61%, so m = 0.61 against 0.13 on the Svanberg curve

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The gap, 1.62 points of GDP, splits exactly, in logs, between the two factors that differ: exposure 14% → 18% and profitability 30% → 45%

Behind profitability: exposed-occupation labor costs 45% of the US level in Bahamas and 8% in Bolivia, while the AI system costs about the same in both

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Across the 22 countries exposure ranges 14% to 20% and profitability 25% to 45%. Cheaper AI narrows the profitability gap (it was 9% to 19% against 23% at 2023 costs), but it remains the main difference

The labor share multiplies TFP and cancels in GDP (appendix A16) →
Where the workers go: the trends, not AI (appendix A13) →

3.3%

ROBUSTNESS

At 2023 costs: 20% to 29% of the benchmark; scale effects: +0.3 to +0.5 pp of GDP growth

COUNTRY · GDP GAIN AFTER TEN YEARS, %	(1) BENCHMARK: AGHION-BUNEL CALIBRATION (COSTS −22% A YEAR, A = 40%)	(2) ACEMOGLU CALIBRATION (2023 COSTS, A = 27%)	(3) NO COST GAP WITH THE US (Q = 50%)	(4) SMALLER SAVING ON HARD-TO-LEARN TASKS (ACEMOGLU EQ. 15)	(5) MODEL GDP, CES CAPITAL-LABOR (Σ_{KL} = 0.7)	(6) EXTENSION: WITH SCALE EFFECTS THROUGH IDEAS (B = 3.1)
0.9%	3.6%	2.6%	·	3.8%		
Barbados	**3.1%**	0.8%	3.8%	2.5%	·	3.5%
Costa Rica	**2.8%**	0.7%	3.7%	2.2%	·	3.2%
Argentina	**2.8%**	0.6%	4.0%	2.2%	2.3%	3.1%
Uruguay	**2.7%**	0.7%	3.6%	2.2%	2.4%	3.1%
Belize	**2.6%**	0.6%	3.6%	2.0%	·	3.0%
Chile	**2.4%**	**0.5%**	**3.3%**	**1.9%**	**2.0%**	**2.7%**
Brazil	**2.3%**	0.5%	3.5%	1.8%	2.0%	2.6%
Jamaica	**2.2%**	0.4%	3.7%	1.8%	·	2.6%
Mexico	**2.2%**	0.5%	3.4%	1.7%	1.8%	2.7%
Panama	**2.1%**	0.5%	3.0%	1.7%	1.7%	2.7%
Colombia	**2.1%**	0.4%	3.5%	1.7%	·	2.5%
El Salvador	**2.1%**	0.4%	3.4%	1.7%	1.8%	2.5%
Dominican Republic	**2.1%**	0.4%	3.4%	1.6%	1.7%	2.5%
Guatemala	**2.1%**	0.4%	3.3%	1.6%	1.7%	2.5%
Peru	**2.0%**	0.4%	3.2%	1.6%	1.5%	2.4%
Guyana	**2.0%**	0.4%	3.0%	1.5%	·	2.3%
Honduras	**1.9%**	0.4%	3.2%	1.5%	1.6%	2.3%
Suriname	**1.9%**	0.4%	3.8%	1.5%	·	2.3%
Paraguay	**1.8%**	0.4%	3.2%	1.5%	·	2.2%
Ecuador	**1.8%**	0.4%	2.9%	1.4%	1.5%	2.1%
Bolivia	**1.6%**	0.3%	2.8%	1.3%	1.3%	2.0%
United States*	**3.7%**	1.2%	3.7%	3.0%	·	4.1%

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Uruguay	**2.7%**	0.7%	3.6%	2.2%	2.4%	3.1%
Belize	**2.6%**	0.6%	3.6%	2.0%	·	3.0%
Chile	**2.4%**	**0.5%**	**3.3%**	**1.9%**	**2.0%**	**2.7%**
Brazil	**2.3%**	0.5%	3.5%	1.8%	2.0%	2.6%
Jamaica	**2.2%**	0.4%	3.7%	1.8%	·	2.6%
Mexico	**2.2%**	0.5%	3.4%	1.7%	1.8%	2.7%
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Colombia	**2.1%**	0.4%	3.5%	1.7%	·	2.5%
El Salvador	**2.1%**	0.4%	3.4%	1.7%	1.8%	2.5%
Dominican Republic	**2.1%**	0.4%	3.4%	1.6%	1.7%	2.5%
Guatemala	**2.1%**	0.4%	3.3%	1.6%	1.7%	2.5%
Peru	**2.0%**	0.4%	3.2%	1.6%	1.5%	2.4%
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United States*	**3.7%**	1.2%	3.7%	3.0%	·	4.1%

(2) Acemoglu's calibration (profitability at 2023 costs, 9% to 19%; a = 27%): 0.3% to 0.9% of GDP. The pace at which AI gets cheaper and better is the assumption that matters

(4) Hard-to-learn tasks. Acemoglu (eq. 15) separates tasks without a clear measure of success, where AI is harder to train: 28% of exposed tasks. On those the saving is cut from 40% to 10%, so the bound falls by about 21%, to 1.3% to 2.6%

(5) CES capital-labor (σ_{KL} = 0.7): capital responds less to TFP; model GDP 12% to 24% below the bound (1% to 11% with Cobb-Douglas)

(6) Extension: scale effects. If AI also makes research cheaper, more ideas are produced: +0.2% of TFP in ten years, +1.3% in the long run. GDP gain 2.0% to 3.8%, +0.3 to +0.5 pp over the benchmark (appendix A17 →)

* United States with Acemoglu's exposure of 20%: (1) and (3) are 20% × q 50% × a 40%; (2) is Acemoglu's own number. Aghion and Bunel's own US number uses an exposure of 60%: 12% of GDP in ten years (6.8% of TFP, 0.7 pp a year); (3) gives every country the US profitability. (6) Semi-endogenous growth (Jones 1995; Bloom, Jones, Van Reenen and Webb 2020): ideas get harder to find at rate β; frontier ideas assumed to reach every country. Aghion in detail: appendix A14 →

What this means for Latin America

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CONCLUSIONS

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6. Replicable with public data

ILOSTAT and CEPALSTAT are enough; easy to update as adoption costs change

> **These numbers may sound small; the technology will still be very disruptive**

Thank you

federico.huneeus@duke.edu

Interactive calculator with all the numbers, by country and scenario

fedehuneeus.com/s/research/ai_latinamerica_dashboard.html

17.4% of Chile's wage bill is in exposed tasks; US: 20.0%



Same task-level exposure for each occupation; the weights are Chile's wage bill by occupation (CASEN 2022 household survey, 4-digit ISCO-08)

The gap is the occupational mix. Technicians are 12.5% of the wage bill in Chile and 18.6% in the US; clerks 5.2% versus 7.8%

Robust to the source. The ILO index (Gmyrek et al. 2025) puts Chile between 15.2% and 18.6%. A Chilean task study (CENIA-SOFOFA 2024) and PIAAC 2023 confirm the translation holds

A different technology than the Central Bank's. The September 2026 IPoM automation-risk measure ranks occupation groups the opposite way (rank correlation −0.62): traditional automation hits manual work, generative AI hits office work. A11

In Chile only 12.9% of exposed tasks pay off, versus 23% in the US

Cost of the AI system

Only in-house engineers get cheaper; data, compute and the external vendor are paid in dollars



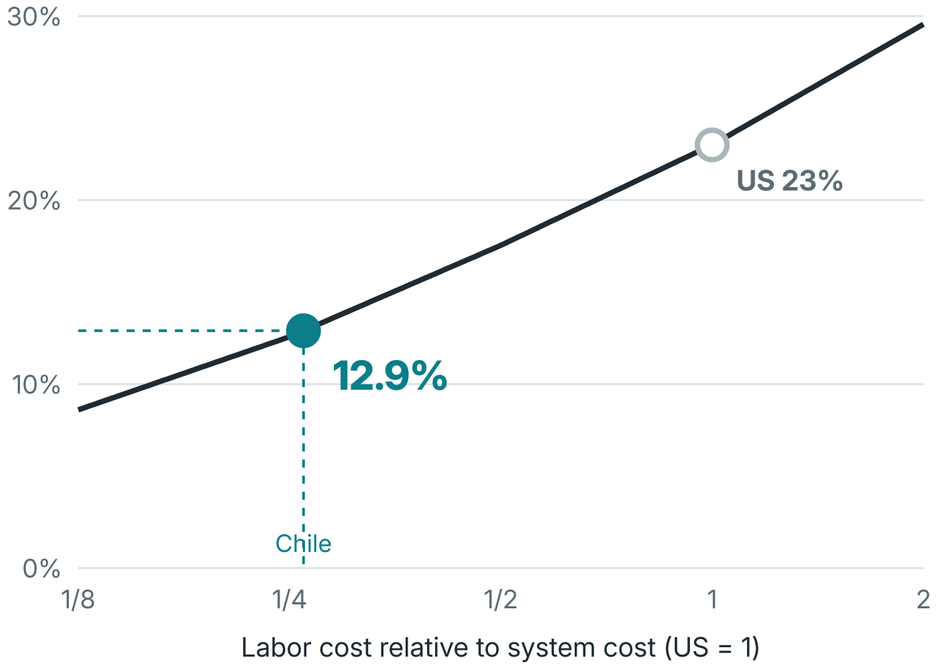
Data and compute, in dollars · External vendor, in dollars · In-house engineers

Cost of the labor it replaces

Chilean wage plus employer contributions (INE labor-cost index), relative to the US



Exposed tasks worth automating (Svanberg et al. 2024, Fig. 6b)



Labor falls to 17% of the US cost, the system only to 66%. The ratio, **0.26**, read on the Svanberg et al. curve gives **12.9%**

Half the US effect, and eroding slowly as sectoral weights move

(A) WITH TODAY'S WEIGHTS (2022)

0.32% TFP

0.60% GDP

in ten years; 0.25% and 0.48% with hard tasks. United States: 0.66% and 1.16%

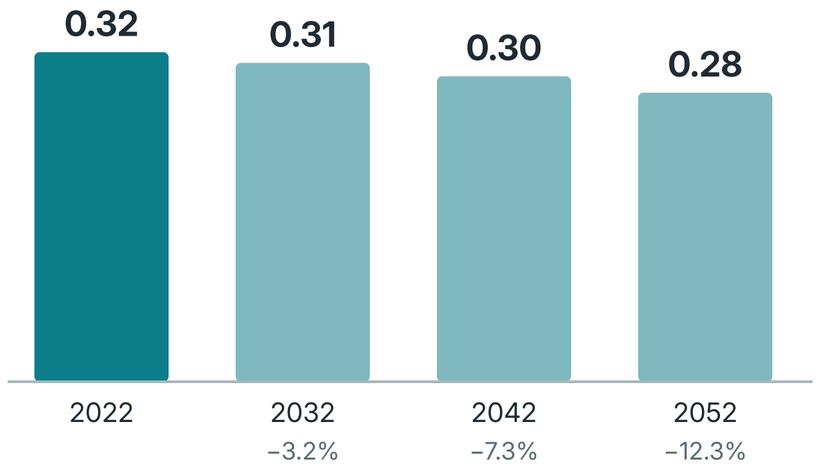
(B) WITH THE WEIGHTS OF 2032

0.31% TFP

0.57% GDP

−3.2% relative to 2022 weights: the activities AI touches lose GDP weight (39.5% → 39.4%) even as they gain workers (46.5% → 47.5%)

TFP gain of the same AI wave, by the year of the GDP weights, %



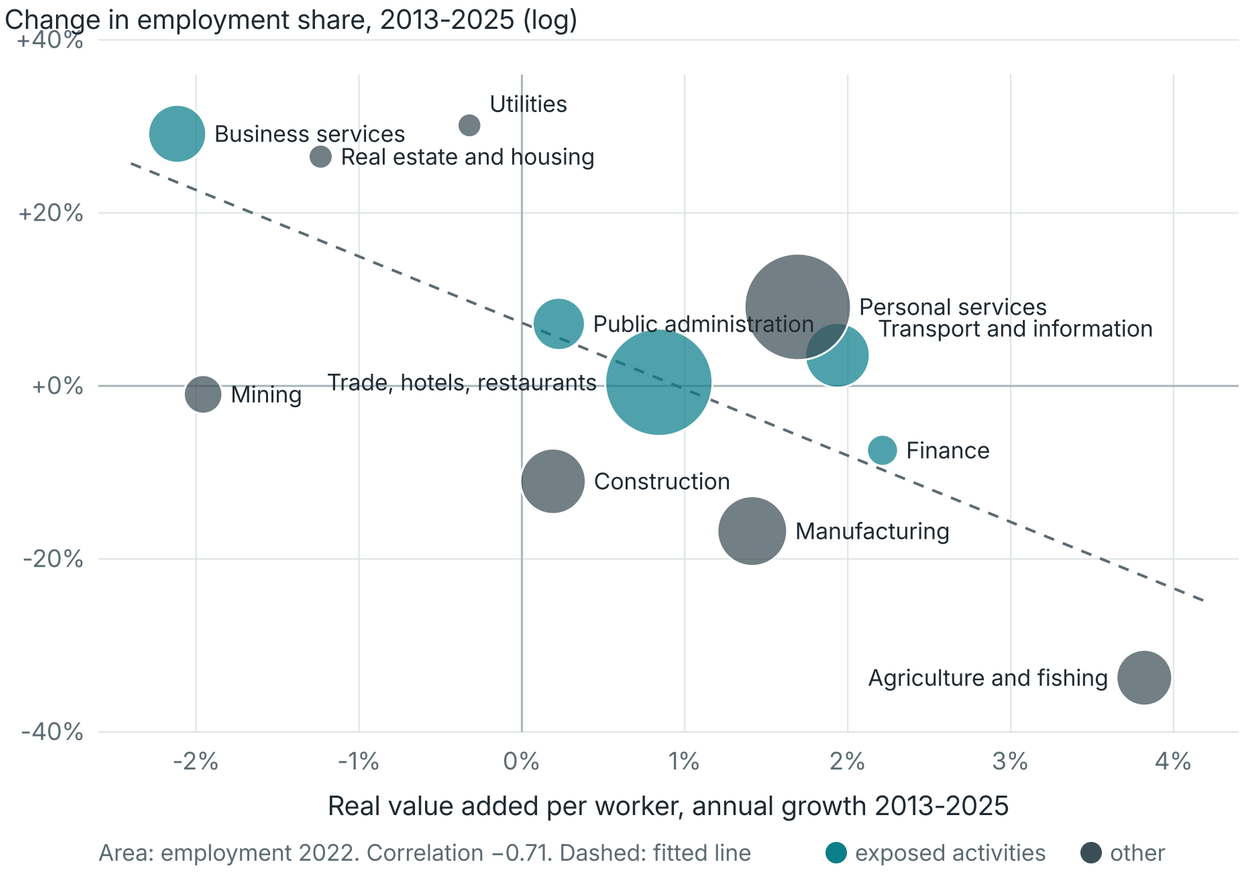
The same AI wave buys a bit less each decade: 0.30% with 2042 weights, 0.28% with 2052 weights (−12.3%). The exact GDP effect in the model is 0.55% against 0.59% from the formula (the income effect); 0.50% with CES capital-labor (multiplier 1.60 instead of 1.85, robustness)

Profitability explains almost all of the gap with the US



Each bar replaces one US factor with its Chilean value, in this order. In any of the 24 orders, profitability is by far the largest bar (between −0.44 and −0.55 points). Without separating hard tasks

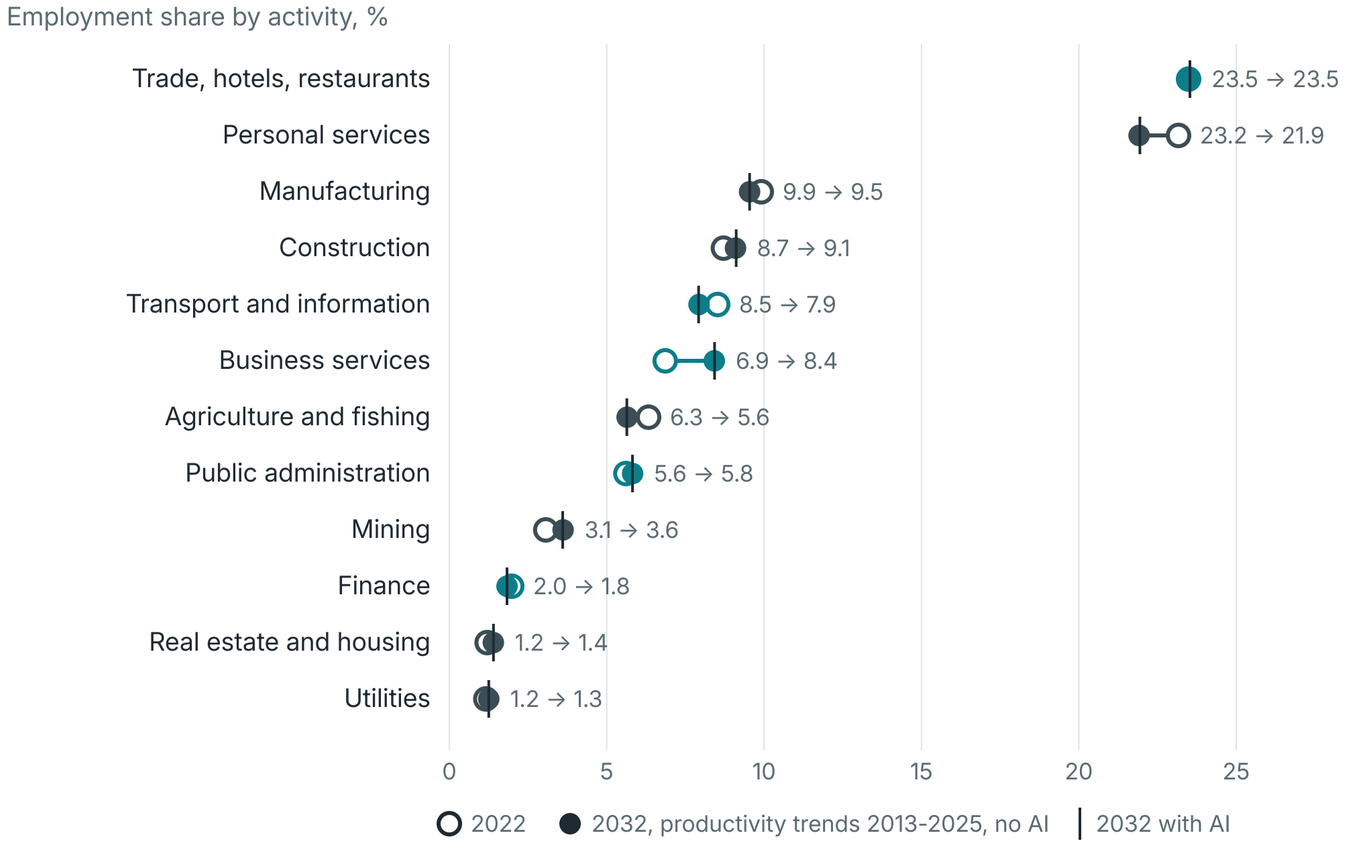
Chile 2013-2025: faster productivity, falling employment share



- Twelve activities of the Central Bank's national accounts; employment from the INE survey via ILOSTAT.
- Correlation −0.71: employment moves to the activities where productivity lags
- Colored: the exposed activities (trade, transport and information, finance, business services, public administration)

Validation, 2013-2025. Fed with observed productivity growth by activity, the model predicts employment-share changes with correlation 0.70 and slope 1.05; without the agricultural income effect, 0.47

Productivity trends move employment across sectors; AI does not



Colored: exposed activities (exposure of their workers ≥ 0.20). Trend reallocation 3.1 pp; AI 0.06 pp

Between 2022 and 2032 the productivity trends move **3.1 points** of employment across activities, toward the lagging ones (construction, business services, mining). AI moves **0.06 points**

> A tenfold larger AI shock would move only **0.6 points** across sectors: AI's employment effects are within sectors (tasks), not between them

What would change Chile's number

VARIANT	TASKS WORTH AUTOMATING	TFP, 10 YEARS	GDP, 10 YEARS
Main estimate (Chilean wages and system cost)	**12.9%**	**0.32%**	**0.60%**
Separating hard-to-learn tasks (7% saving)	12.9%	0.25%	**0.48%**
Exposure from the ILO index (anchored, two variants)	12.9%	0.28% to 0.34%	**0.53% to 0.64%**
All engineering at Chilean wages	18.3%	0.45%	**0.86%**
Labor cost +100% (severance, rigidities; outside the method)	17.9%	0.44%	**0.84%**
The paper's 23% (US adoption economics)	23.0%	0.57%	**1.07%**
Acemoglu's US bound	23.0%	0.66%	**1.16%**

> Exposure is robust across sources; what moves the result is the cost of adopting AI. The Baumol erosion is proportional, so it applies to every row

The structural-transformation block under other parameters

CASE	σ	TREND WINDOW	TFP, 2022 WEIGHTS	TFP, 2032 WEIGHTS	CHANGE	EMPLOYMENT MOVED BY AI, PP	BY TRENDS, PP
Benchmark: H-R nest, CLM elasticities, Cobb-Douglas production	**0.35**	**2013-2025**	**0.318%**	**0.308%**	**−3.2%**	**0.056**	**3.05**
Other trend window	0.35	2013-2019	0.318%	0.315%	**−1.1%**	0.055	3.24
Other trend window	0.35	2019-2025	0.318%	0.298%	**−6.3%**	0.059	4.56
Other elasticity	0.50	2013-2025	0.318%	0.311%	**−2.4%**	0.044	2.44
Other elasticity	0.80	2013-2025	0.318%	0.316%	**−0.8%**	0.020	1.25
Other elasticity	1.00	2013-2025	0.318%	0.320%	**0.3%**	0.009	0.87
Other elasticity	1.50	2013-2025	0.318%	0.329%	**3.2%**	0.044	2.68
Hard tasks, benchmark	0.35	2013-2025	0.253%	0.245%	**−3.2%**	0.044	3.05
Paper's 23%, benchmark	0.35	2013-2025	0.568%	0.549%	**−3.2%**	0.101	3.05
Homothetic non-agricultural aggregate	0.35	2013-2025	0.318%	0.307%	**−3.6%**	0.050	3.01
Single homothetic CES over the 12 sectors	0.35	2013-2025	0.318%	0.307%	**−3.5%**	0.048	3.22
No agricultural subsistence term ($\bar{c}$ = 0)	0.35	2013-2025	0.318%	0.307%	**−3.6%**	0.052	2.97
Comin-Lashkari-Mestieri σ = 0.84	0.84	2013-2025	0.318%	0.317%	**−0.5%**	0.017	1.09
Satiation: exposed activities ε = 0.6 (assumption)	0.35	2013-2025	0.318%	0.306%	**−3.8%**	0.122	2.97
CES capital-labor, σ_KL = 0.7 (Oberfield-Raval 2021): model GDP 0.50% vs 0.55% Cobb-Douglas	0.35	2013-2025	0.318%	0.308%	**−3.2%**	0.053	3.05
CES σ_KL = 0.5 (Gechert et al. 2022): model GDP 0.47%	0.35	2013-2025	0.318%	0.308%	**−3.2%**	0.053	3.04
CES σ_KL = 1.25 (Karabarbounis-Neiman 2014): model GDP 0.59%	0.35	2013-2025	0.318%	0.308%	**−3.2%**	0.062	3.05

The accounting and the Baumol block, country by country

COUNTRY	EXPOSURE E	WAGE / US, EXPOSED	PROFIT. Q	LABOR SHARE	TFP BOUND	GDP BOUND	BAUMOL CORR.	VALIDATION CORR.	EROSION 2032
Bahamas	18.1%	0.56	45.0%	0.34	1.12%	**3.27%**	·	·	·
Barbados	18.8%	0.38	41.9%	0.55	1.74%	**3.15%**	·	·	·
Costa Rica	18.5%	0.23	38.3%	0.55	1.57%	**2.83%**	·	·	·
Argentina	20.2%	0.13	34.3%	0.50	1.39%	**2.77%**	−0.51	0.37	−0.2%
Uruguay	17.8%	0.25	38.5%	0.56	1.53%	**2.74%**	−0.55	0.56	−0.2%
Belize	18.0%	0.19	35.8%	0.46	1.18%	**2.57%**	·	·	·
Chile	**16.7%**	**0.19**	**35.5%**	**0.54**	**1.29%**	**2.38%**	**−0.64**	**0.66**	**−2.7%**
Brazil	17.4%	0.12	33.1%	0.58	1.33%	**2.30%**	−0.70	0.69	1.9%
Jamaica	18.3%	0.11	30.2%	0.54	1.19%	**2.21%**	·	·	·
Mexico	16.8%	0.12	32.2%	0.36	0.79%	**2.16%**	−0.94	0.90	−5.9%
Panama	15.0%	0.18	35.6%	0.35	0.75%	**2.14%**	−0.94	0.94	−2.0%
Colombia	17.3%	0.10	30.8%	0.51	1.08%	**2.13%**	·	·	·
El Salvador	16.8%	0.11	31.3%	0.51	1.07%	**2.11%**	−0.68	0.66	−2.9%
Dominican Republic	17.2%	0.10	30.2%	0.48	0.99%	**2.08%**	−0.85	0.89	0.7%
Guatemala	16.6%	0.11	31.0%	0.44	0.90%	**2.06%**	−0.88	0.87	−2.2%
Peru	16.2%	0.10	30.6%	0.43	0.85%	**1.98%**	−0.76	0.78	2.6%
Guyana	14.9%	0.13	32.9%	0.55	1.08%	**1.96%**	·	·	·
Honduras	16.2%	0.10	30.1%	0.57	1.12%	**1.95%**	−0.83	0.88	−1.8%
Suriname	19.1%	0.06	25.4%	0.50	0.97%	**1.94%**	·	·	·
Paraguay	15.8%	0.08	29.1%	0.45	0.83%	**1.84%**	·	·	·
Ecuador	14.4%	0.11	31.2%	0.57	1.03%	**1.80%**	−0.48	0.48	−4.9%
Bolivia	13.8%	0.09	29.9%	0.53	0.87%	**1.65%**	−0.74	0.72	−2.7%

Sources: ILOSTAT (employment by occupation at 2 digits of ISCO-08, mean monthly earnings of employees by occupation in US dollars, labor income share), CEPALSTAT (value added by activity at current and constant prices), exposure of Eloundou et al. (2024) by occupation. Employer surcharge: Chile's measured value scaled by each country's statutory rate. Erosion with the structural-transformation block (σ = 0.35, Huneeus-Rogerson nest, Comin-Lashkari-Mestieri income elasticities); · = not computed. Venezuela, Nicaragua, Trinidad and Tobago, Haiti, Cuba: no dollar earnings or recent accounts

The paper's numbers across versions

VERSION	TFP BOUND	TFP, HARD TASKS	GDP	GDP, HARD TASKS
Working paper, MIT, April 2024	0.71%	0.55%	1.1%	0.92% (0.90% updated)
NBER WP 32487, May 2024	0.66%	0.53%	1.16%	0.93%
Economic Policy 40(121), 2025 (published; used here)	0.66%	0.53%	1.16% (1.4–1.56% with the full capital response)	0.93%
SSRN entry, revised October 8, 2026	0.66%	0.53%	same abstract as the published version	
Interview, Fortune, June 21, 2026	about 0.55%		1 to 1.5%; about 5% of tasks automated	
NBER WPs 2026 (34854 pro-worker AI; 34910 knowledge collapse; 35036 AI aggregation)	no new macro estimates			

The calculation uses the published version. No later version with new estimates exists as of November 2026; the June 2026 interview restates the published bound

AI exposure versus the Central Bank's automation risk

OCCUPATION GROUP (ISCO 1 DIGIT)	AUTOMATION RISK, MEAN (IPOM)	HIGH-RISK SHARE (IPOM)	AI EXPOSURE, MEAN (THIS STUDY)	SHARE ABOVE MEDIAN EXPOSURE
Managers	0.40	20%	**0.15**	80%
Professionals	0.36	15%	**0.17**	51%
Technicians	0.41	22%	**0.29**	79%
Clerical support	0.43	51%	**0.44**	100%
Services and sales	0.53	62%	**0.23**	80%
Agricultural workers	0.63	97%	**0.07**	0%
Crafts	0.55	85%	**0.04**	3%
Machine operators	0.58	99%	**0.12**	56%
Elementary occupations	0.61	99%	**0.06**	17%

Private salaried workers with a written contract, as in the IPoM box (Albagli et al. 2026, Tables 8 and 9): their automation risk is the Nedelkoska-Quintini (2018) probability by occupation applied to administrative records; our AI exposure is Eloundou et al. (2024) by occupation on CASEN 2022. Rank correlation across the nine groups −0.62, across twelve sectors −0.33

How the model works, step by step

1 · Sectors and data

Agriculture plus 11 non-agricultural sectors (ISIC sections), with value added at constant and current prices (CEPALSTAT) and employment (ILOSTAT), per country

2 · Preferences

Huneeus-Rogerson nest: log utility over agriculture, with a subsistence level $\bar{c}_a$ (income effect), and the non-agricultural aggregate C. C is the nonhomothetic CES of Comin, Lashkari and Mestieri: one elasticity of substitution $\sigma = 0.35$ (from Huneeus and Rogerson) and their income elasticities ε_i by sector (manufacturing = 1)

3 · Production

Cobb-Douglas in labor and capital in each sector, with the sector's labor share; labor mobile across sectors at one wage; capital rented at a fixed real price, so the capital-output ratio is constant (Acemoglu's assumption for the GDP multiplier). Robustness: CES capital-labor with $\sigma_{KL} = 0.7$ (Oberfield and Raval 2021; 0.5 and 1.25 in A8), where the capital response to TFP is $1 + \sigma_{KL}\, s_K/s_L$ instead of $1/s_L$

4 · Trends, no AI

Each sector's productivity grows at its observed 2013-2025 rate (longest window available); the equilibrium is solved year by year for 10, 20 and 30 years. In-sample validation over the same window: predicted vs actual changes in employment shares correlate 0.37 to 0.94

5 · AI on top

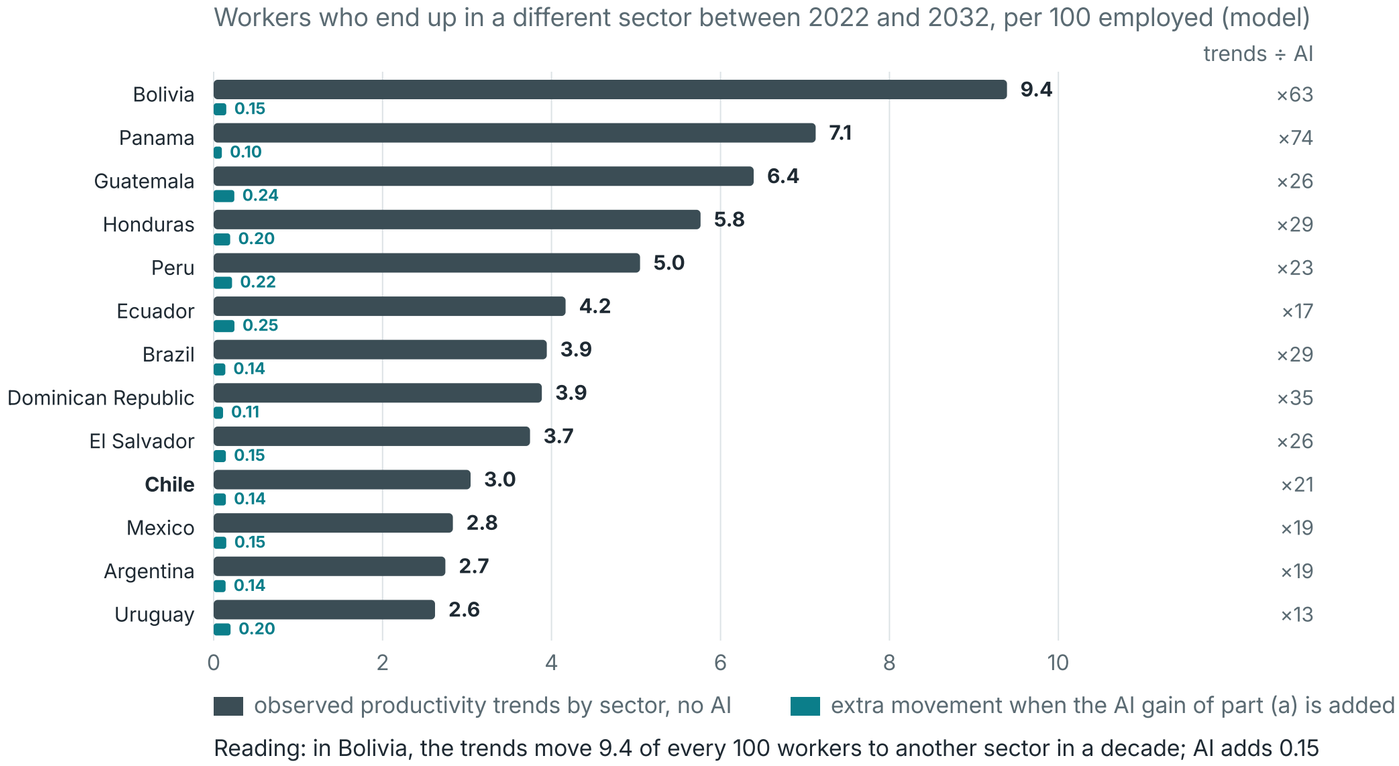
The part (a) gain enters each sector as an extra productivity increase: exposure of its workers × q × a × the sector's labor share, phased over ten years, on top of the trends. Each sector gets a different shock

6 · Outputs

Sectoral GDP and employment shares with and without AI; the Hulten gain $\Sigma\, s_i \times \text{gain}_i$ with the 2022, 2032 and 2042 weights (part (b)); the exact GDP effect in the model (2.20% for Chile vs 2.38% from the formula; 1.96% with CES); workers moved across sectors

← back to the results

Productivity trends move workers across sectors; AI does not



Reading: in Bolivia, the trends move 9.4 of every 100 workers to another sector in a decade; AI adds 0.15

How to read it. For each country, the share of workers who end up in a different sector between 2022 and 2032 in the model: with the observed productivity trends alone (dark), and the extra movement when the AI gain of part (a) is added on top (teal)

A result, not an assumption. The AI gain is 0.7% to 1.7% of TFP over a decade, spread over several sectors. With complements ($\sigma = 0.35$) a sector's nominal weight moves by roughly $(1 - \sigma)$ times its relative productivity gain, so the shift is at most 0.25 of every 100 workers. The trends are many times larger and cumulate

Why it matters. At these magnitudes AI reshapes tasks within occupations, not the sectoral structure; the Baumol drift of part (b) comes from the trends. Ten times Acemoglu's shock would move 0.6 of every 100 workers in Chile (appendix A6)

← back to robustness

Same mechanism as Aghion; here it is measured, and slow

Aghion, Jones and Jones (2019)

- Tasks are gross complements (σ < 1): GDP is limited by its **weak links**, the tasks not yet automated
- Automated goods get cheaper and **lose GDP share**: automation and Baumol's cost disease are the same force
- Result: growth is set by what is essential and hard to improve; even near-complete automation leaves the labor share near two thirds
- An **asymptotic** argument: it binds as the automated fraction approaches one

Aghion and Bunel (2024)

- Acemoglu's formula with a different reading of each term: median **0.68 pp a year** of TFP growth over a decade (range 0.07 to 1.24), against Acemoglu's 0.07
- The gap: exposure 60% against 20%, profitability 50% against 23%, cost saving 40% against 27%. The benchmark here takes their q and a, not their exposure
- No Baumol term; the electricity analogy gives 0.8 to 1.3 pp a year
- OECD (Filippucci, Gal and Schief 2024): direct effect + input-output multiplier − a Baumol reallocation drag, 0.25 to 0.6 pp a year for the US

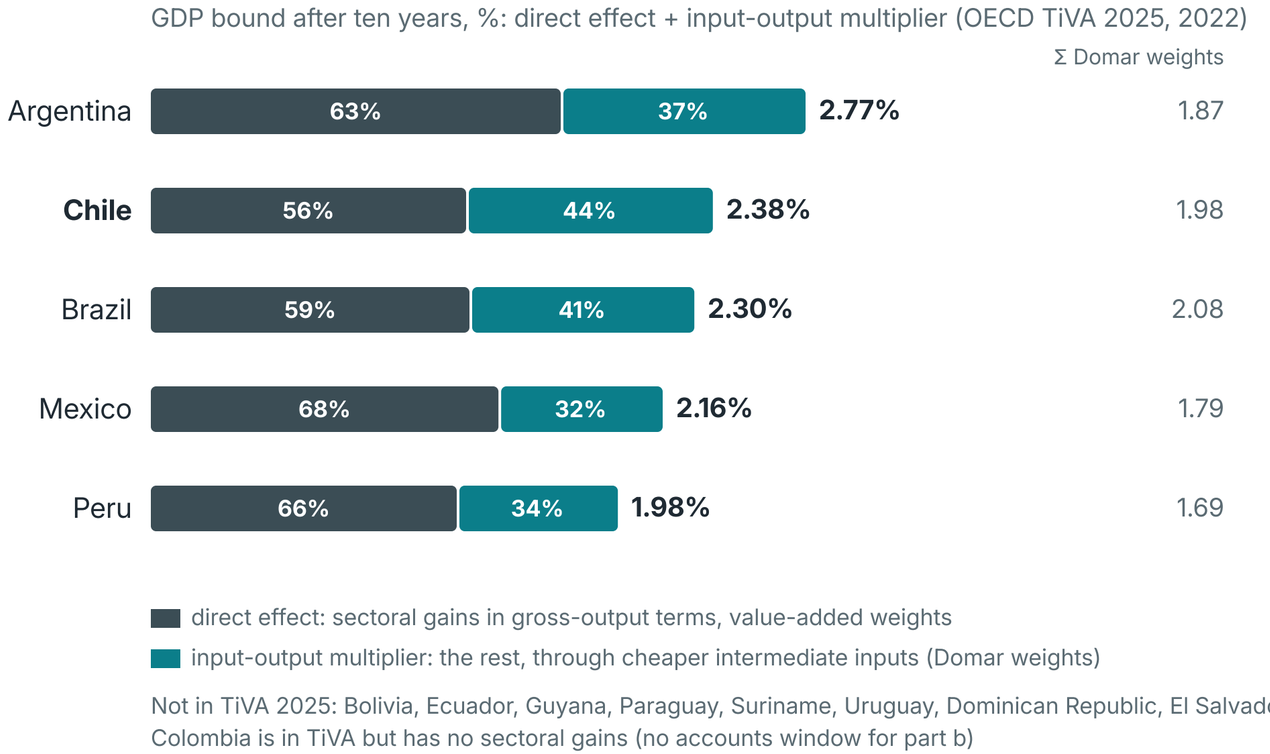
This paper

- The same σ < 1 force, run on **observed sectoral trends** for 13 countries: the weights drift a few points per decade
- The AI gain changes by −0.49 to +0.07 pp of GDP over twenty years (−22.5% to 3.7% of the gain)
- The relative erosion does not depend on the size of the shock: with Acemoglu's calibration it is the same share of a gain 3.5 to 5.0 times smaller
- The weak links are the laggard sectors that gain weight (health, education, construction, trade in some countries): they carry small AI gains, which is what caps the aggregate

Agreement on the mechanism, disagreement on the horizon: Aghion's bottleneck binds as automation nears completion; at a 0.7% to 1.7% TFP shock it is a drift of at most 0.49 pp of GDP, and it is the trends, not AI, that move the weights

← back to robustness

Input-output links: already inside the bound, and 32% to 44% of it



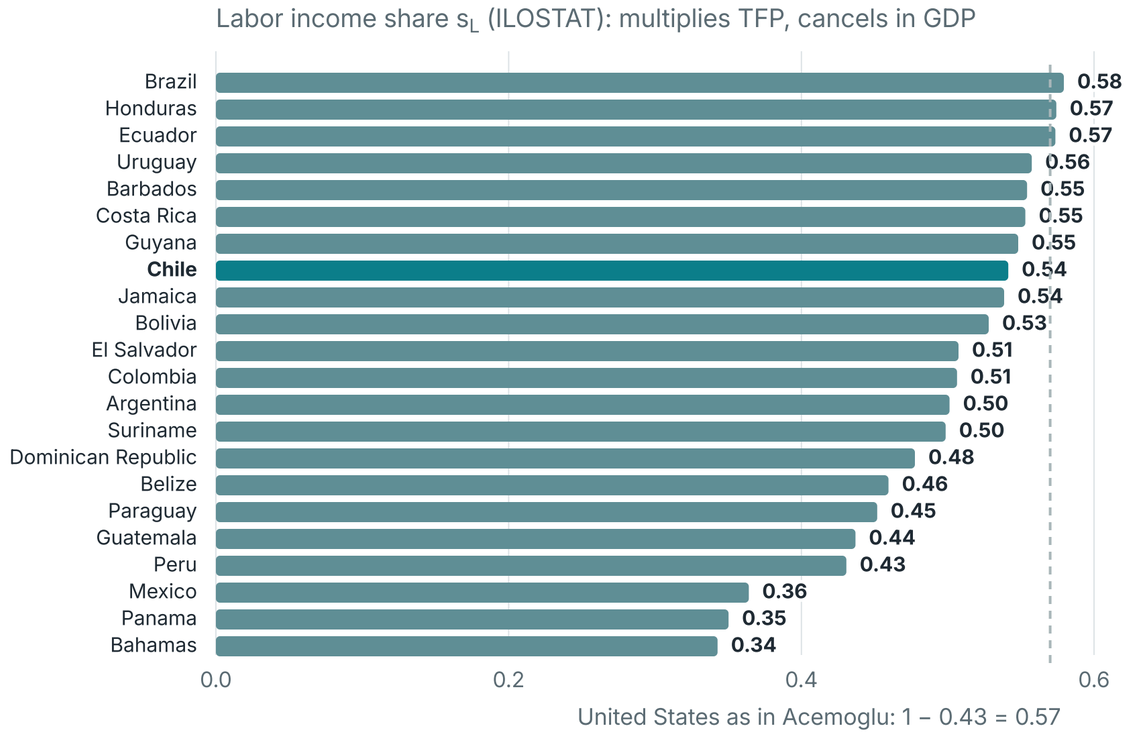
The OECD decomposition. A sector's gain in value-added terms, g_i, becomes a gross-output gain $g_i \times VA_i/GO_i$. The **direct effect** weights it by value-added shares; the **input-output multiplier** is the Domar-weighted sum (GO_i/GDP, which add to 1.7 to 2.1) minus the direct effect

Why it adds up to the bound. $(GO_i/GDP) \times (VA_i/GO_i) \times g_i = s_i \times g_i$: with labor-cost savings as the shock, Hulten's value-added accounting already contains the pass-through of cheaper inputs. The exercise shows how much of the bound travels through supply chains, not an extra gain

Data. OECD TiVA 2025 (ICIO) covers Argentina, Brazil, Chile, Colombia, Costa Rica, Mexico, Peru among the 22 countries, so this is robustness, not the benchmark. With TiVA's own sectoral weights the Domar-weighted total is 87% to 100% of the CEPALSTAT-weighted bound (classification differences, mainly finance and real estate). Beyond first order, Filippucci et al. find that general-equilibrium reallocation lowers gains when demand is inelastic, the Baumol channel of part (b)

← back to the results

The labor share multiplies TFP and cancels in GDP



TFP = e × q × a × s_L. The labor share turns the wage-bill share of exposed tasks into a GDP share: 0.34 (Bahamas) to 0.58 (Brazil)

GDP = TFP / s_L. Capital follows TFP at a constant capital-output ratio, so the same share divides back out: GDP = e × q × a

What it changes. The TFP ranking, not the GDP ranking: a low labor share lowers the TFP bound and raises the capital multiplier by the same factor. ILOSTAT imputes the labor part of self-employment income, with different treatment across countries

← back to robustness

If AI also makes research cheaper: +0.18% of TFP in ten years

Ideas. New ideas need research input S and get harder to find as the stock A grows: growth of $A = \theta S^{\lambda} A^{-\beta}$ (Jones 1995; Aghion, Jones and Jones 2019). $\beta = 3.1$ for the US economy with $\lambda = 1$ (Bloom, Jones, Van Reenen and Webb 2020, Table 7)

AI. Research is made of tasks; AI makes the exposed ones cheaper, as in part (a). Each dollar of research buys $\Delta = -\ln(1 - e \times q \times a) = 4.1\%$ more effective input, at the frontier (e 20%, q 50%, a 40%)

Result. A level effect, not permanent growth: TFP ends $\lambda\Delta/\beta = 1.3\%$ higher. It arrives slowly, 3.1% of the remaining gap a year ($\beta \times$ 1% TFP growth, Jones 2022): 14% of it in ten years. GDP = TFP / s_L, on top of the bound

SCENARIO	RESEARCH INPUT, Δ	TFP, 10 YEARS	TFP, 20 YEARS	TFP, LONG RUN	GDP, 10 YEARS, REGION
Benchmark of the extension: β = 3.1, λ = 1, research as exposed as the average US task (20%), adoption phased in over ten years	**+4.1%**	**+0.18%**	**+0.49%**	**+1.3%**	**+0.3 to +0.5 pp**
Research tasks more exposed: 40%	+8.3%	**+0.38%**	+0.99%	+2.7%	+0.7 to +1.1 pp
Ideas get harder to find more slowly: β = 2	+4.1%	**+0.19%**	+0.53%	+2.0%	+0.3 to +0.6 pp
Ideas get harder to find faster: β = 5	+4.1%	**+0.17%**	+0.43%	+0.8%	+0.3 to +0.5 pp
Stepping on toes in research: λ = 0.75, β = 2.4 (their Table A1)	+4.1%	**+0.14%**	+0.38%	+1.3%	+0.2 to +0.4 pp
No phase-in: AI fully adopted in research from the first year	+4.1%	**+0.35%**	+0.61%	+1.3%	+0.6 to +1.0 pp

An exploration, not part of the bound. It assumes frontier ideas reach every country one for one and at once (an upper bound for the region), frontier TFP growth of 1% a year, and research tasks no more exposed than the average task. Product variety, new tasks and automation of idea production itself (which could raise growth permanently) are outside it